Vashu Tyagi

Data Analyst

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Summary

Data Analyst & MCA Candidate with hands-on experience in Python, SQL, and transactional database architectures (MySQL). Proven track record in automating data pipelines, optimizing complex query execution, and building cloud-ready analytics engines. Specialized in converting unstructured datasets into actionable business metrics using RFM segmentation, cohort retention modeling, statistical analysis, and interactive Power BI dashboards.

Skills

Core Analytics & Concepts: Relational Data Modeling, RFM Segmentation, Cohort Retention, Automated ETL Pipelines, Data Cleansing, Data Validation, Statistical Analysis
Programming & Querying: Python (Pandas, NumPy, SQLAlchemy), SQL / MySQL (CTEs, Window Functions, Joins, Aggregations)
Dashboards & Visualizations: Power BI (DAX, Data Modeling), Streamlit, Advanced Excel (Pivot Tables, Lookup Functions)
Developer Tools & Environments: Git, GitHub, WSL (Windows Subsystem for Linux), Linux CLI

Experience

Data Analysis Intern **June 2026 – Present**
Palin Analytics (On-Site) *Tools: Python, Pandas, MySQL, Power BI, WSL*

- Engineered automated Python/Pandas processing pipelines to clean and restructure **9,000+ multi-source transactional records**, reducing manual data processing latency by **40%** and eliminating data anomalies.
- Designed and optimized complex MySQL queries utilizing multi-table joins and window functions, cutting analytical reporting query execution time by **30%**.
- Developed interactive Streamlit web apps and real-time tracking dashboards for executive stakeholders, cutting recurring manual reporting effort by **50%**.
- Leveraged WSL (Windows Subsystem for Linux) CLI workflows and Linux utility tools to execute development workflows, version control, and data pipeline management.

Projects

Customer RFM & Cohort Retention Analytics Engine | *SQL, Power BI, DAX, Python, Pandas* **GitHub Repository**

- Architected an end-to-end analytics pipeline processing **1,067,371 transactional records** across **5,878 unique customers** using SQL CTEs and window functions for dynamic 1–5 RFM scoring.
- Designed multi-page executive Power BI dashboards with DAX measures to expose user decay trends, pinpointing a critical **65% onboarding drop-off at M+1**.
- Validated Pareto revenue concentration proving that top **20% Champions generate 76.8% of total revenue** (£13.63M out of £17.74M total monetary volume).

Education

Master of Computer Applications (MCA) **July 2026 – Present**
Amity University (Online) *Gurugram, India*

Bachelor of Computer Applications (BCA) **Completed: April 2026**
Maa Shakumbhari University *Muzaffarnagar, India*